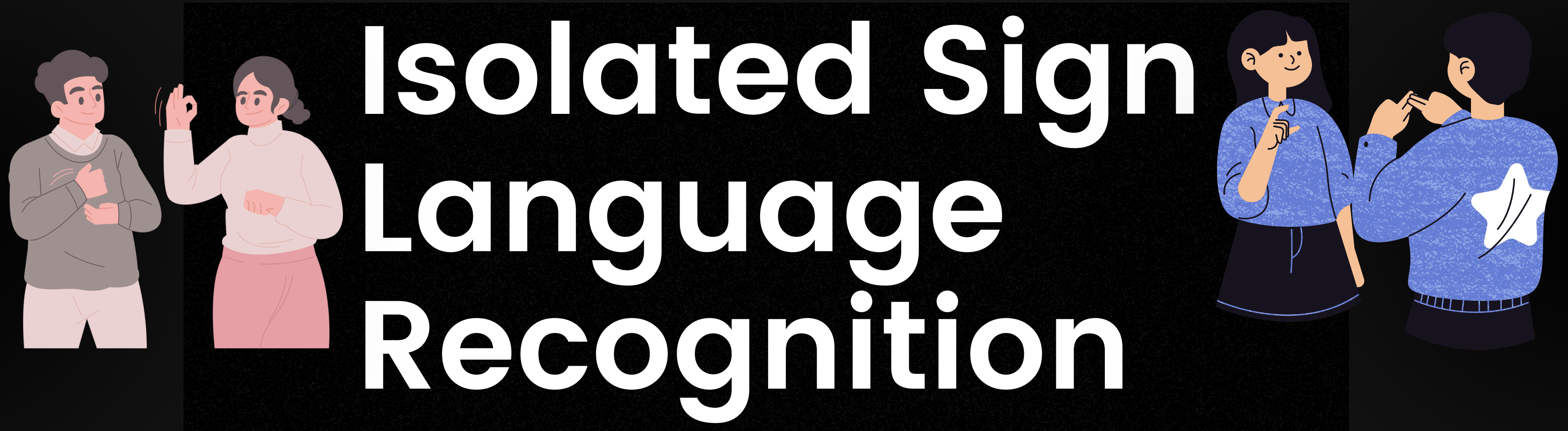




Isolated Sign Language Recognition

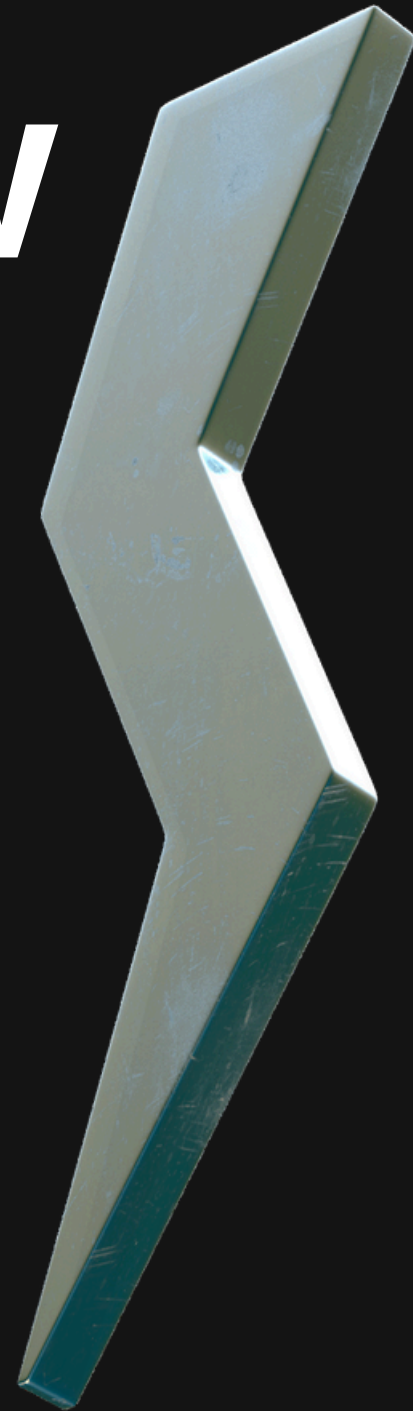
What it is and why it's important

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-Mentor: Prof. Mrugendra Rahevar

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Overview

What you need to know



- Isolated Sign Language Recognition using AI and Machine Learning.
- Goal: Build a model to recognize isolated signs from American Sign Language (ASL) vocabulary.
- Dataset: Isolated Sign Language Recognition Corpus (version 1.0).
- Supported Tasks: Sign Language Recognition.

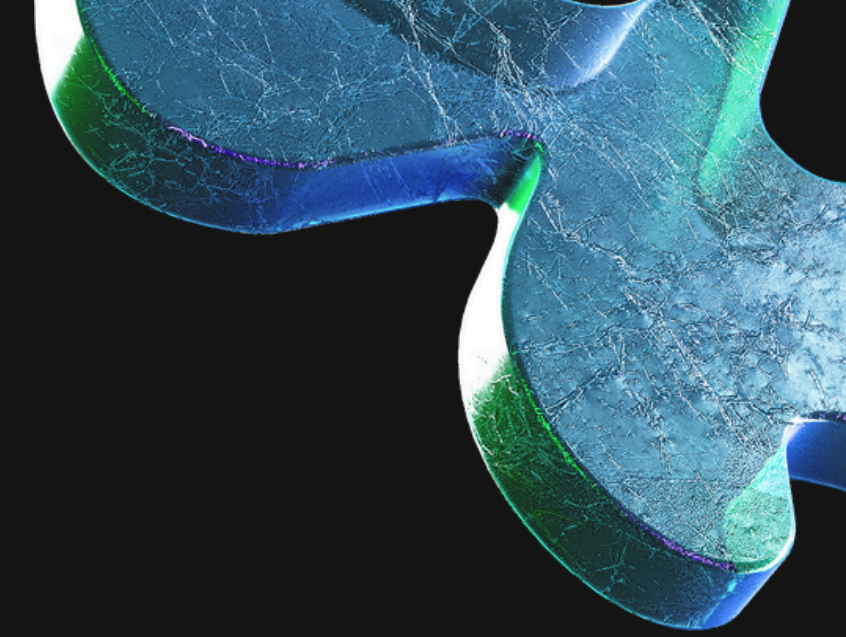
Dataset Summary

- The Isolated Sign Language Recognition corpus (version 1.0) details:
- Hand and facial landmarks from **Mediapipe** on ~100k videos of isolated signs.
- 21 Deaf signers participated with a **250-sign vocabulary**.
- Language: American Sign Language.

Dataset Structure:

- Data Instances: Example data field with 'frame', 'type', 'landmark_index', 'x', 'y', 'z'.
- Data Fields: Detailed data field information available on Kaggle.

Dataset Creation and Curation



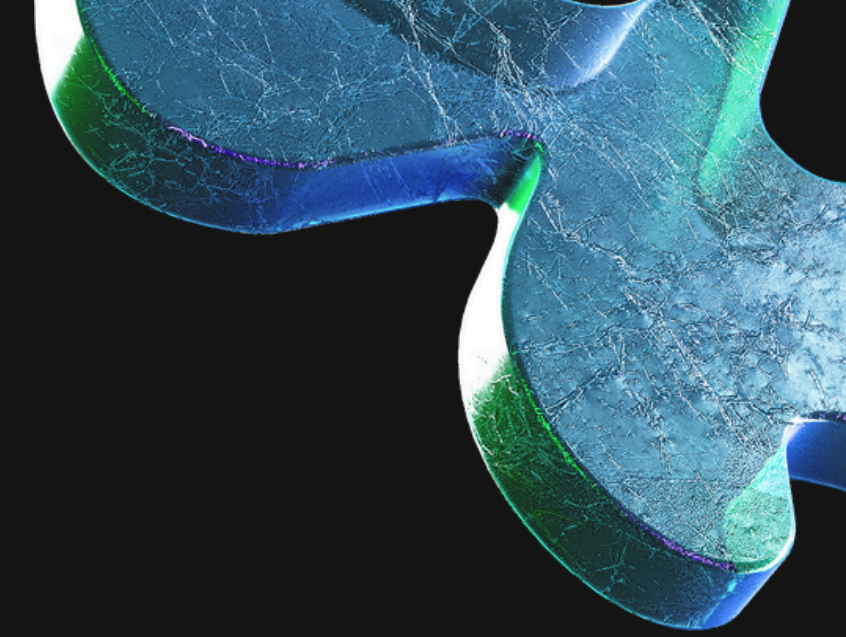
Curation Rationale:

- Dataset represents 250 first concepts taught to infants in any language.
- Aim to create isolated sign recognizer for educational games to help hearing parents learn ASL.
- Importance of avoiding Language Deprivation Syndrome in Deaf children.

Source Data:

- 21 signers recruited by the Deaf Professional Arts Network.
- Randomly selected English concepts to sign with smartphones.

Data Collection and Normalization



Data Collection Process:

- Signers recorded videos while signing each concept.
- Video extraction with buffer 0.5 seconds before button press and after button release.
- Variants of signs due to background, region, fingerspelling, and more.

Data Normalisation:

- Handling left and right-handed signers, switching signing hands.
- Recognition system to handle variations.

Model Training Script Overview

DATA PREPROCESSING

Model Architecture

MODEL COMPILATION AND
TRAINING

Model Evaluation and
Conversion

Data Preprocessing

- Loading the Isolated Sign Language Recognition Corpus dataset.
 - Preprocessing Steps:
 - Extracting relevant hand and facial landmarks using Mediapipe.
 - Creating a DataFrame of landmarks for each frame and sign type (face, pose, left hand, right hand).



Model Architecture



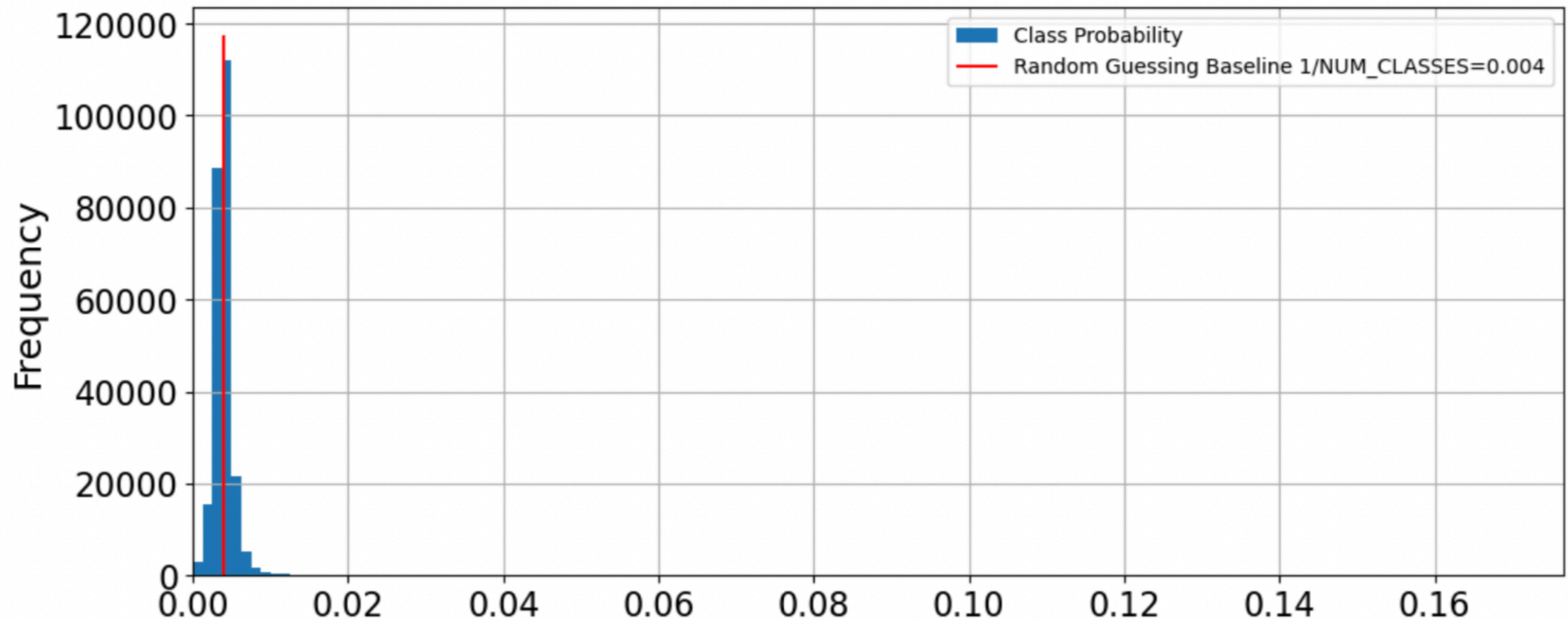
Model Overview:

- Using TensorFlow and Keras to build the Isolated Sign Language Recognition model.

Architecture Details:

- Input and masking
- Extract lips, hands, and pose landmarks
- Flatten layer
- Transformer
- Dense layers for classification.
- Softmax activation for multi-class classification.

Softmax Output Initialized Model | $\mu=0.004$, $\sigma=0.002$



Model Compilation and Training

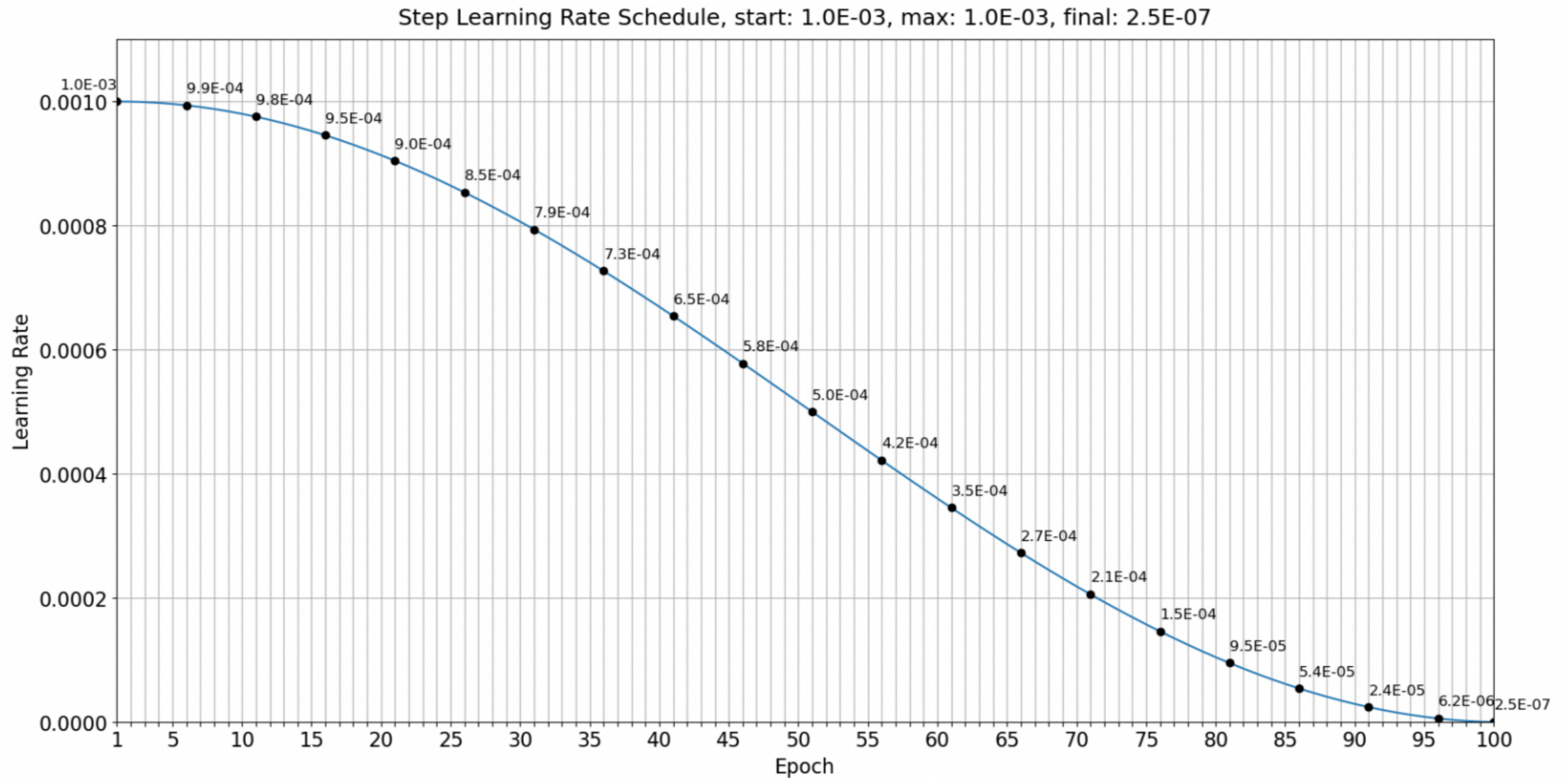


Model Compilation:

- Configuring the model for training.
- Setting the loss function (Sparse Categorical Crossentropy) and optimizer (Adam).
- Choosing evaluation metrics (accuracy).

Model Training:

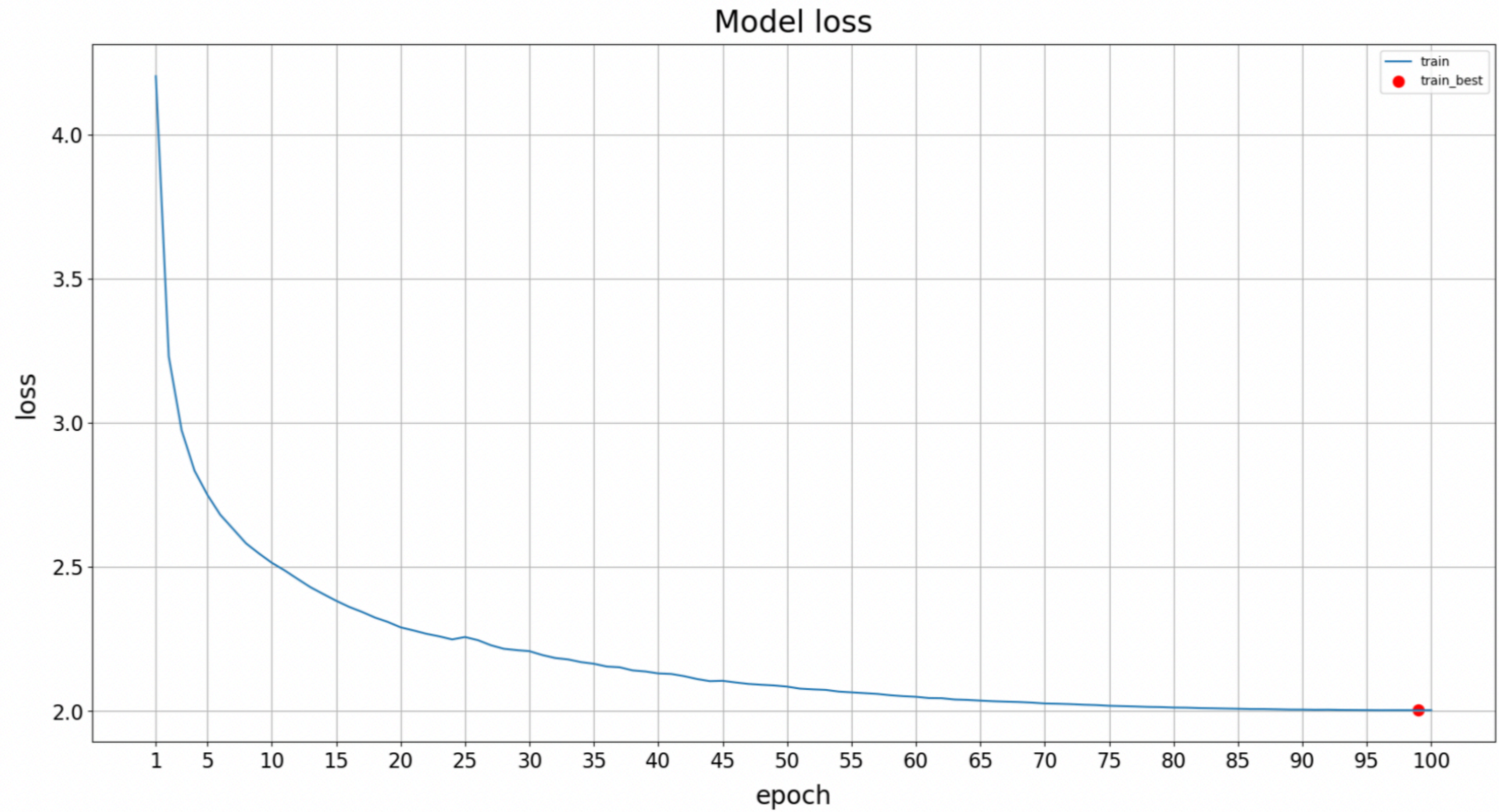
- Splitting the dataset into training and validation sets.
- Training the model using the training data.
- Monitoring training progress with validation data.

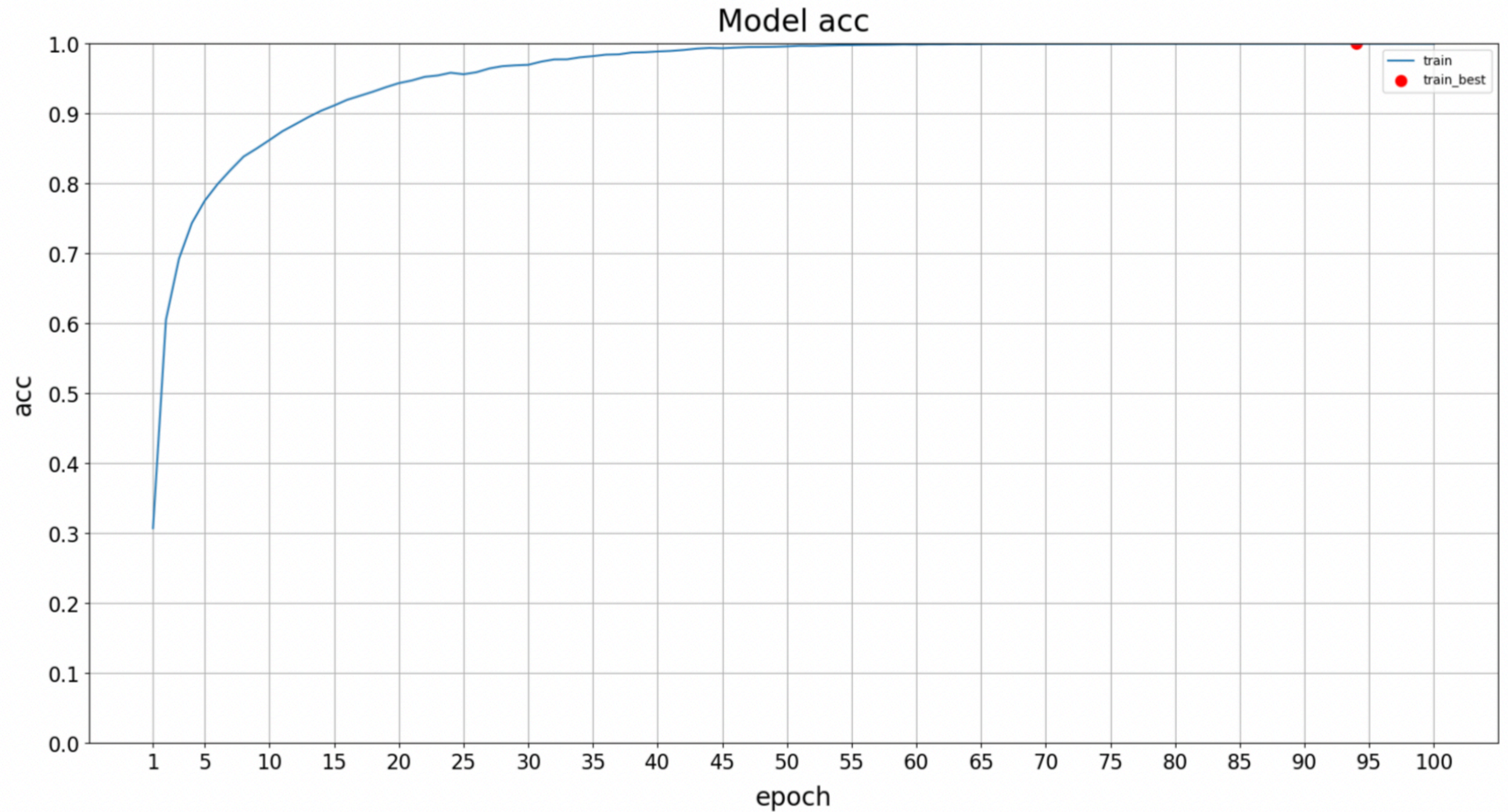


Model Evaluation and Conversion

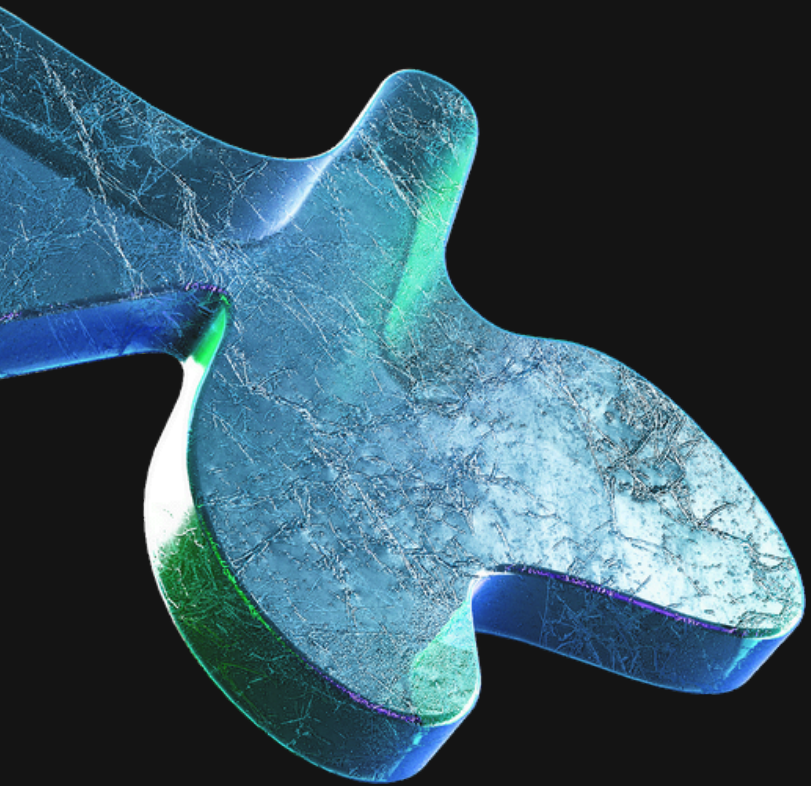
- Model Evaluation:
 - Evaluating the model performance on the validation set.
 - Calculating accuracy and other relevant metrics.
- Model Conversion:
 - Converting the trained Keras model to TensorFlow Lite format.
 - Saving the TFLite model to a file.







App Implementation Overview



App Implementation



- The Isolated Sign Language Recognition app allows users to capture sign language gestures and obtain real-time predictions.
- The app uses the Mediapipe library to extract hand and facial landmarks from the user's video feed.
- The captured landmarks are processed and used as input to the trained TensorFlow Lite model for prediction.
- The predicted sign is then displayed to the user along with its confidence score.

App Architecture



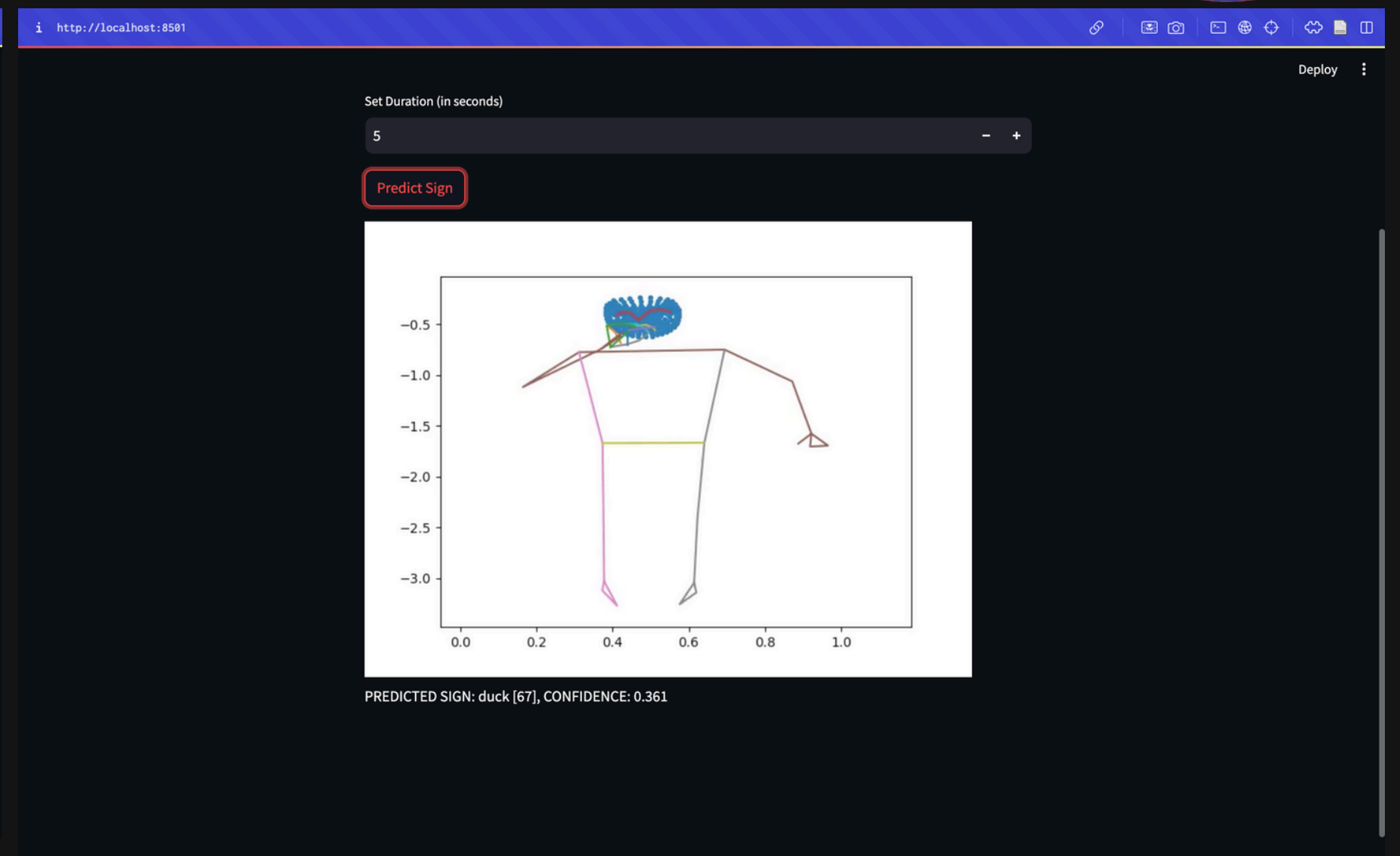
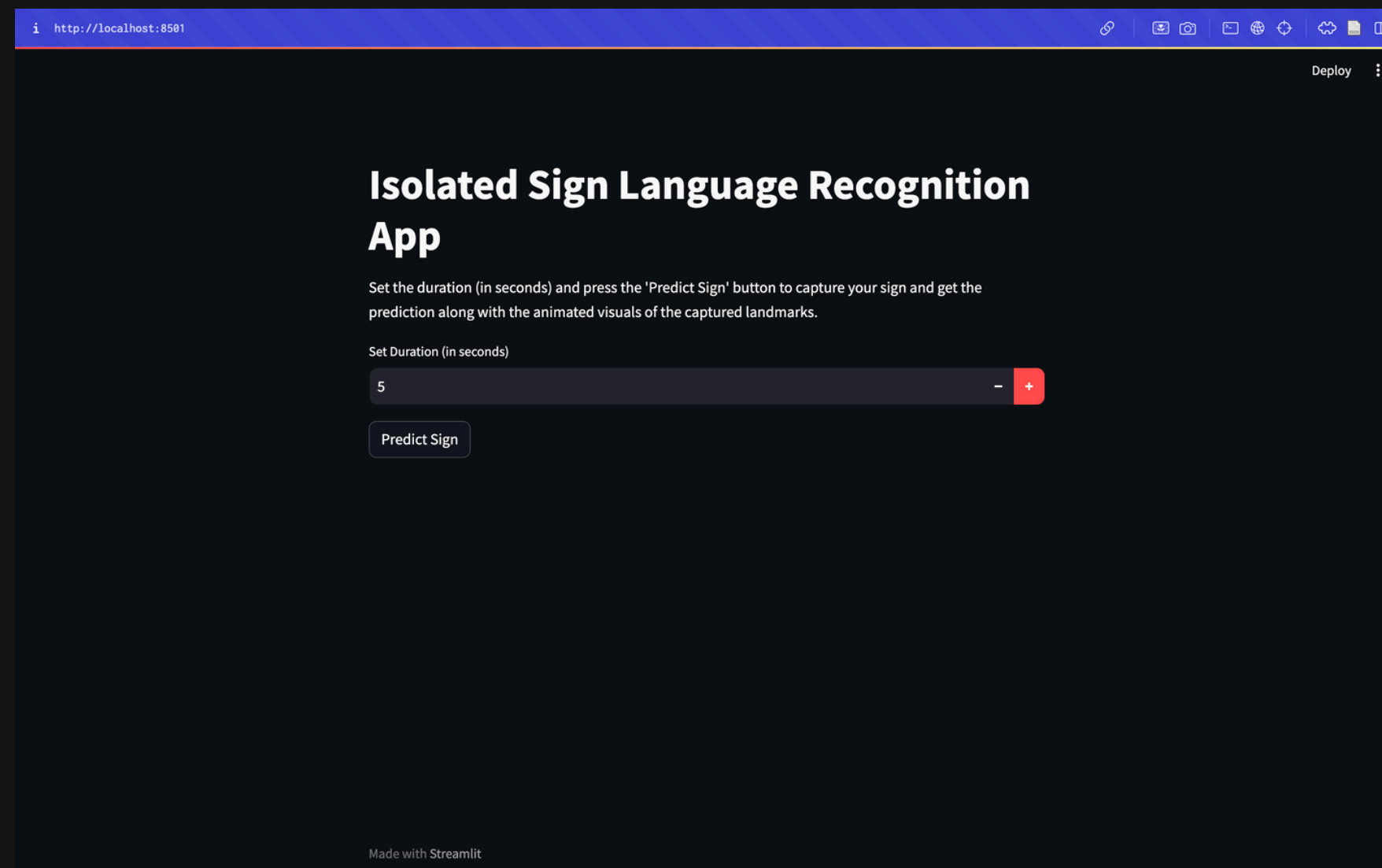
- The app is built using the Streamlit framework, which allows for easy integration of machine learning models into web applications.
- Key Components:
- Video Capture: The app uses the user's webcam to capture real-time video.
- Landmark Extraction: Mediapipe is used to extract hand and facial landmarks from each video frame.
- Model Inference: The landmarks are fed into the TensorFlow Lite model to obtain predictions.
- Display: The predicted sign and its confidence score are displayed to the user.

App Demonstration



- The app is simple to use:
- The user sets the duration (in seconds) for capturing the sign gesture.
- After clicking the "Start Capture" button, the webcam captures video for the specified duration.
- The landmarks are extracted using Mediapipe, and the video is processed frame by frame.
- Once the video capture is complete, the captured landmarks are passed to the TensorFlow Lite model for prediction.
- The predicted sign and its confidence score are displayed to the user

App Demonstration



Conclusion

- Isolated Sign Language Recognition is an essential technology for educational games and tools that aim to teach ASL to hearing parents of Deaf children.
- The developed model and app serve as valuable tools to bridge the communication gap between Deaf and hearing individuals.
- The project contributes to providing an inclusive learning environment for Deaf children and raising awareness about Deaf culture.



The background of the slide is a dark, abstract composition. On the right side, there is a large, swirling, liquid-like shape in shades of blue and purple, resembling a stylized flower or a complex fluid pattern. This shape has a glossy, reflective texture with highlights and shadows. The rest of the background is a solid dark color, possibly black or very dark blue.

Thank You!